UNIVERSIDAD NACIONAL DE TRUJILLO

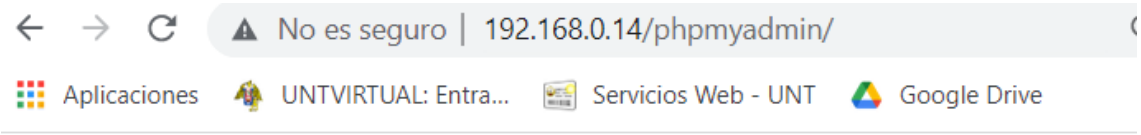
Dr. Ing. Gómez Avila José

# LABORATORIO 08

**Objetivo**: Al finalizar el estudiante experimenta en un entorno controlado el manejo de servidor remoto con los servicios más utilizados en IOT.

1. Requerimientos:
   - Implementaciones realizadas en el laboratorio pasados.
   - Máquina Virtual con Windows.

2. Intentamos ingresar al phpmyadmin del servidor




**Forbidden**

You don't have permission to access this resource.

Additionally, a 403 Forbidden error was encountered while trying to use an ErrorDocument to handle the request.

*Apache/2.4.51 (Win64) OpenSSL/1.1.1l PHP/8.0.12 Server at 192.168.0.14 Port 80*

Debemos corregir nuestros accesos.

3. Desde la máquina anfitrión ingresamos con nuestro usuario “remoto”

```
remoto@WIN-HU5I3SRHA93 C:\xampp\apache\conf\extra>dir/w
 Volume in drive C has no label.
 Volume Serial Number is 1A26-F15D

 Directory of C:\xampp\apache\conf\extra

[.]                         [..]                        httpd-ajp.conf
httpd-autoindex.conf        httpd-dav.conf              httpd-default.conf
httpd-info.conf             httpd-languages.conf        httpd-manual.conf
httpd-mpm.conf              httpd-multilang-errordoc.conf  httpd-proxy.conf
httpd-ssl.conf              httpd-userdir.conf          httpd-vhosts.conf
httpd-xampp.conf            proxy-html.conf
              15 File(s)         45,825 bytes
               2 Dir(s)  29,417,697,280 bytes free

remoto@WIN-HU5I3SRHA93 C:\xampp\apache\conf\extra>
```

```
remoto@WIN-HU5I3SRHA93 C:\xampp\apache\conf\extra>nano httpd-xampp.conf
```

Editar la línea asociada a la ruta de phpmyadmin

```
  </Directory>

  Alias /phpmyadmin "C:/xampp/phpMyAdmin/"
  <Directory "C:/xampp/phpMyAdmin">
      AllowOverride AuthConfig
      Require local
      ErrorDocument 403 /error/XAMPP_FORBIDDEN.html.var
  </Directory>

  Alias /webalizer "C:/xampp/webalizer/"
                          [ Unknown Command ]
^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify  ^C Cur Pos  ^Y Prev Page  M-\ First Line
^X Exit      ^R Read File  ^\ Replace   ^U Uncut Text ^T To Spell ^_ Go To Line ^V Next Page  M-/ Last Line
```

Cambiar la expresión “Require Local” a “Require all granted”. Guardar cambios.

```
    Alias /phpmyadmin "C:/xampp/phpMyAdmin/"
    <Directory "C:/xampp/phpMyAdmin">
        AllowOverride AuthConfig
        Require local
        ErrorDocument 403 /error/XAMPP_FORBIDDEN.html.var
    </Directory>

    Alias /webalizer "C:/xampp/webalizer/"
Search (to replace): Require Local
^G Get Help      M-C Case Sens    M-B Backwards    ^Y First Line    ^P PrevHstory
^C Cancel        M-R Regexp       ^R No Replace    ^V Last Line     ^N NextHstory
```

```
    Alias /phpmyadmin "C:/xampp/phpMyAdmin/"
    <Directory "C:/xampp/phpMyAdmin">
        AllowOverride AuthConfig
        Require local
        ErrorDocument 403 /error/XAMPP_FORBIDDEN.html.var
    </Directory>

    Alias /webalizer "C:/xampp/webalizer/"
Replace with: Require all granted
^G Get Help      ^Y First Line    ^P PrevHstory
^C Cancel        ^V Last Line     ^N NextHstory
```

4. Reiniciar servicio de apache

```
[.]                 [..]                [anonymous]         [apache]            apache_start.bat
apache_stop.bat     catalina_service.bat catalina_start.bat catalina_stop.bat   certificate.crt
[cgi-bin]           [contrib]           ctlscript.bat       [FileZillaFTP]      filezilla_setup.bat
filezilla_start.bat filezilla_stop.bat  [htdocs]            [img]               [install]
killprocess.bat     [licenses]          [locale]            [mailoutput]        [mailtodisk]
[MercuryMail]       mercury_start.bat   mercury_stop.bat    [mysql]             mysql_start.bat
mysql_stop.bat      passwords.txt       [perl]              [php]               [phpMyAdmin]
properties.ini      readme_de.txt       readme_en.txt       [sendmail]          service.exe
setup_xampp.bat     [src]               test_php.bat        [tmp]               [tomcat]
uninstall.dat       uninstall.exe       [webalizer]         [webdav]            xampp-control.exe
xampp-control.ini   xampp-control.log   xampp_shell.bat     xampp_start.exe     xampp_stop.exe
              30 File(s)     16,507,644 bytes
              25 Dir(s)  29,380,513,792 bytes free

remoto@WIN-HU5I3SRHA93 C:\xampp>apache_stop
```

```
remoto@WIN-HU5I3SRHA93 C:\xampp>apache_stop
remoto@WIN-HU5I3SRHA93 C:\xampp>apache_start
Diese Eingabeforderung nicht waehrend des Running beenden
Bitte erst bei einem gewollten Shutdown schliessen
Please close this command only for Shutdown
Apache 2 is starting ...
```

5. Asegurar que el servicio de MySQL este operativo. Luego probamos el ingreso a phpmyadmin y presenta otro impase.



Procedemos a realizar un análisis y vemos cambios a realizar en la configuración de mysql.

6. Buscamos el archivo my.ini y cambiamos los valores de bind-address de 127.0.0.1 a 0.0.0.0

```
mysqlcheck.exe                          mysqld.exe
mysqldump.exe                           mysqldumpslow.pl
mysqld_multi.pl                         mysqlhotcopy.pl
mysqlimport.exe                         mysqlshow.exe
mysqlslap.exe                           mysql_config.pl
mysql_convert_table_format.pl           mysql_install_db.exe
mysql_ldb.exe                           mysql_plugin.exe
mysql_tzinfo_to_sql.exe                 mysql_upgrade.exe
mysql_upgrade_service.exe               mysql_upgrade_wizard.exe
my_print_defaults.exe                   perror.exe
replace.exe                             sst_dump.exe
ucrtbase.dll                            vcruntime140_1.dll
              88 File(s)    142,878,078 bytes
               2 Dir(s)  29,015,494,656 bytes free

remoto@WIN-HU5I3SRHA93 C:\xampp\mysql\bin>nano my.ini
```

Editamos con el mismo procedimiento que se realizó para el paso 3.

```
# Change here for bind listening
bind-address="0.0.0.0"
# bind-address = ::1          # for ipv6

# Where do all the plugins live
plugin_dir="C:/xampp/mysql/lib/plugin/"

# Don't listen on a TCP/IP port at all. This can be a security enhancement,
# if all processes that need to connect to mysqld run on the same host.
# All interaction with mysqld must be made via Unix sockets or named pipes.
# Note that using this option without enabling named pipes on Windows
# (via the "enable-named-pipe" option) will render mysqld useless!
#
# commented in by lampp security
#skip-networking
#skip-federated

# Replication Master Server (default)
# binary logging is required for replication
# log-bin deactivated by default since XAMPP 1.4.11
#log-bin=mysql-bin

# required unique id between 1 and 2^32 - 1
# defaults to 1 if master-host is not set

^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify     ^C Cur Pos
^X Exit        ^R Read File   ^\ Replace     ^U Uncut Text  ^T To Spell    ^_ Go To Line
```

7. Restaurar el servicio de mysql e inicializarlos.

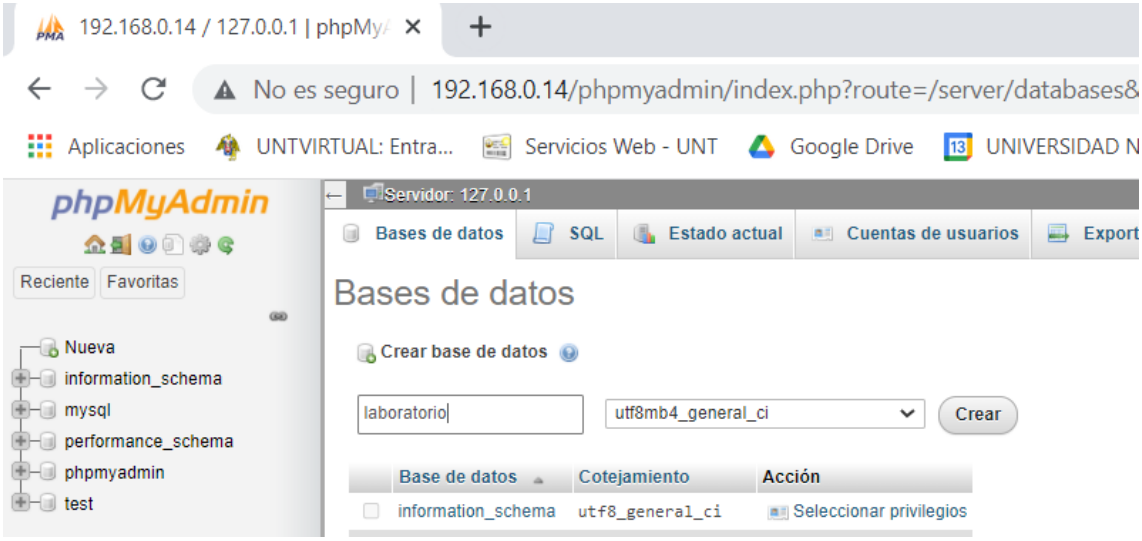
```
remoto@WIN-HU5I3SRHA93 C:\xampp>mysql_stop
Mysql shutdowm ...

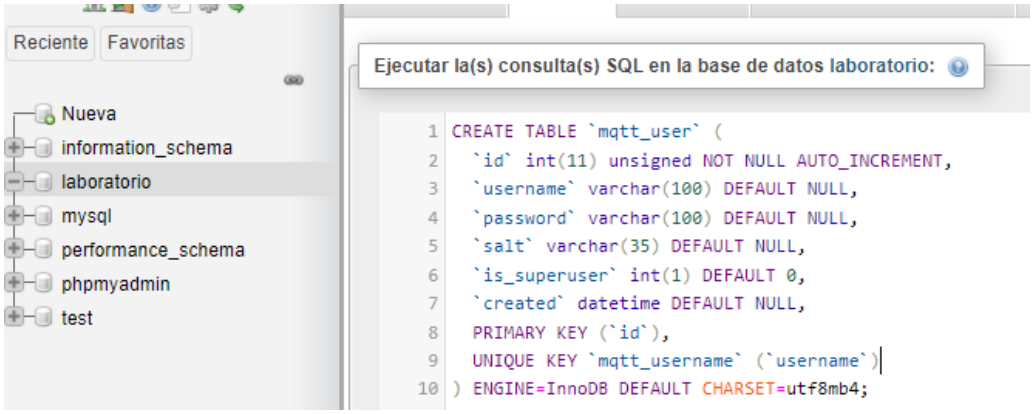
remoto@WIN-HU5I3SRHA93 C:\xampp>mysql_start
Diese Eingabeforderung nicht waehrend des Running beenden
Please dont close Window while MySQL is running
MySQL is trying to start
Please wait  ...
MySQL is starting with mysql\bin\my.ini (console)
2021-12-14  9:26:21 0 [Note] Using unique option prefix 'key_buffer' is error-prone and can break in the fut
use the full name 'key_buffer_size' instead.
2021-12-14  9:26:21 0 [Note] mysql\bin\mysqld (mysqld 10.4.21-MariaDB) starting as process 3660 ...
```

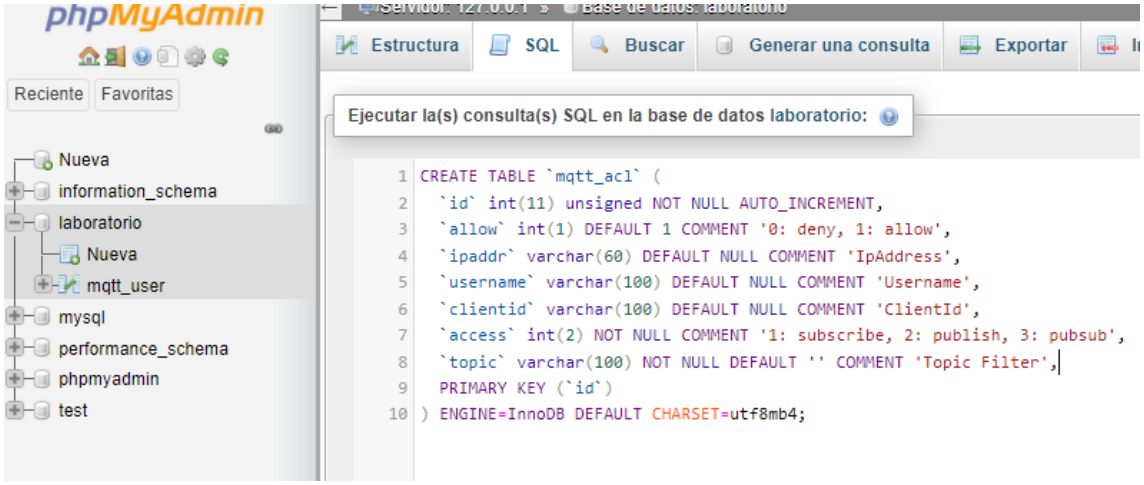
8. Probamos acceso al phpmyadmin



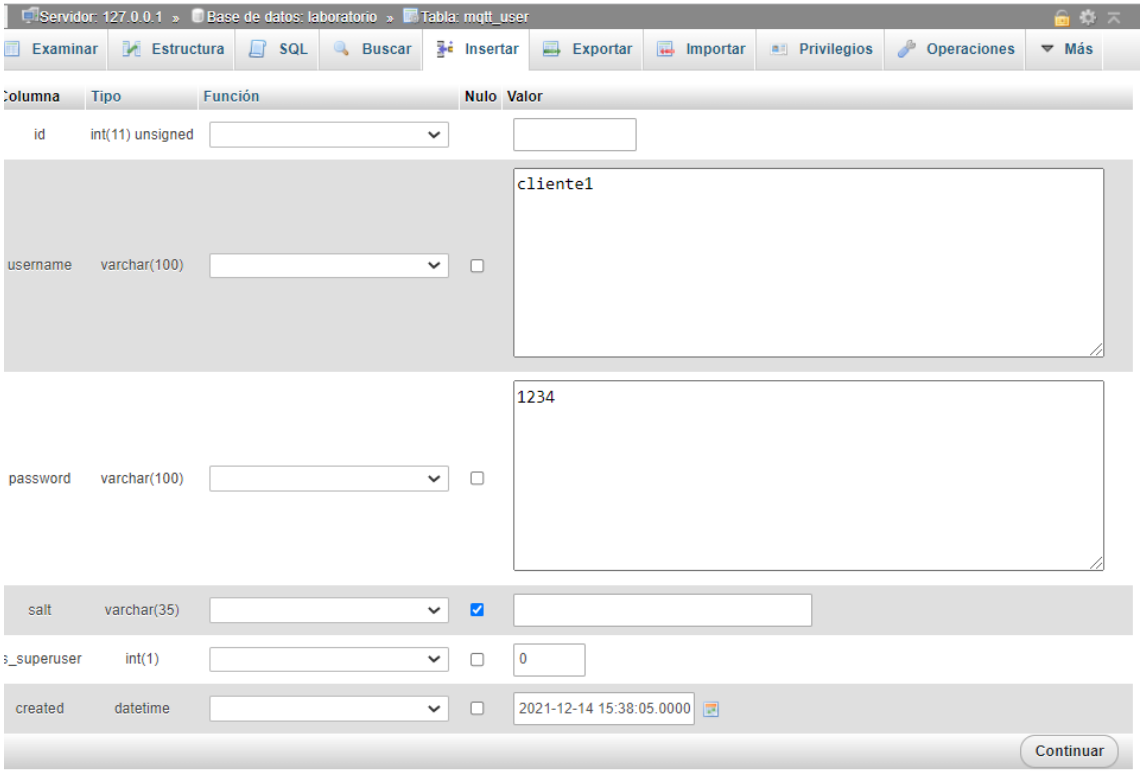

9. Creamos una base de datos nueva llamada “laboratorio”. Luego ubicamos la dirección electrónica: https://docs.emqx.io/en/broker/v4.3/advanced/acl-mysql.html#default-table-structure para crear tablas de mqtt_user y mqtt_acl.



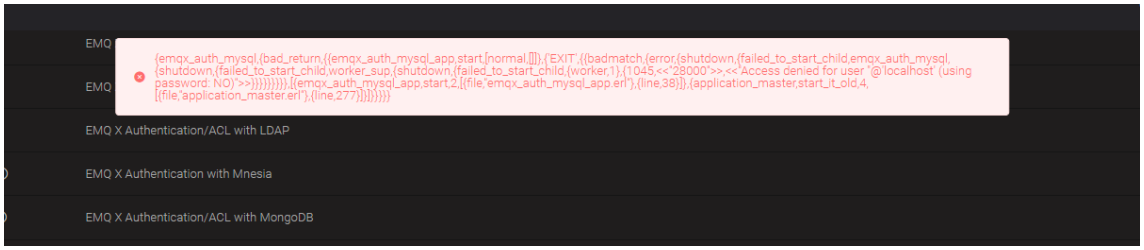



10. Insertamos un usuario.

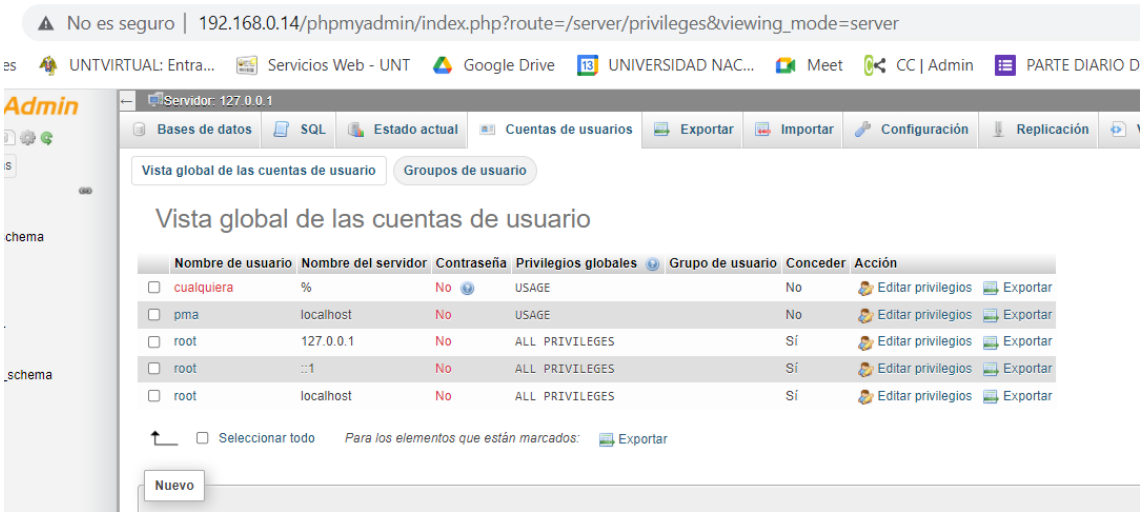


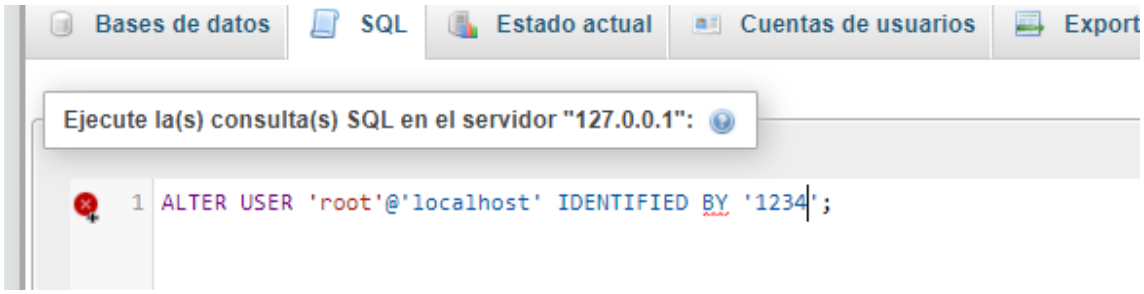

11. Iniciamos el emqx y ubicamos los plugins en especial el de mysql e intentamos iniciar.




Procederemos hacer cambios en la configuración del Mysql y Plugin.

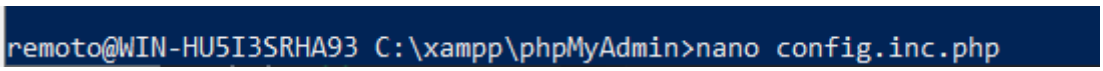
12. Procedemos a agregar clave al superusuario.




La ejecución va implicar cambios en la configuración de phpmyadmin.

13. Regresamos a ubicar el archivo de configuración de phpmyadmin.



Administrator: c:\windows\system32\cmd.exe - nano config.inc.php

```
  GNU nano 2.5.3                        File: config.inc.php

<?php
/*
 * This is needed for cookie based authentication to encrypt password in
 * cookie
 */
$cfg['blowfish_secret'] = 'xampp'; /* YOU SHOULD CHANGE THIS FOR A MORE SECURE COOKIE AUTH! */

/*
 * Servers configuration
 */
$i = 0;

/*
 * First server
 */
$i++;

/* Authentication type and info */
$cfg['Servers'][$i]['auth_type'] = 'config';
$cfg['Servers'][$i]['user'] = 'root';
$cfg['Servers'][$i]['password'] = '';
$cfg['Servers'][$i]['extension'] = 'mysqli';
$cfg['Servers'][$i]['AllowNoPassword'] = true;
$cfg['Lang'] = '';

/* Bind to the localhost ipv4 address and tcp */
$cfg['Servers'][$i]['host'] = '127.0.0.1';
$cfg['Servers'][$i]['connect_type'] = 'tcp';

/* User for advanced features */
$cfg['Servers'][$i]['controluser'] = 'pma';
$cfg['Servers'][$i]['controlpass'] = '';

/* Advanced phpMyAdmin features */
$cfg['Servers'][$i]['pmadb'] = 'phpmyadmin';
$cfg['Servers'][$i]['bookmarktable'] = 'pma__bookmark';
$cfg['Servers'][$i]['relation'] = 'pma__relation';
$cfg['Servers'][$i]['table_info'] = 'pma__table_info';
$cfg['Servers'][$i]['table_coords'] = 'pma__table_coords';
$cfg['Servers'][$i]['pdf_pages'] = 'pma__pdf_pages';
$cfg['Servers'][$i]['column_info'] = 'pma__column_info';
$cfg['Servers'][$i]['history'] = 'pma__history';
$cfg['Servers'][$i]['designer_coords'] = 'pma__designer_coords';
$cfg['Servers'][$i]['tracking'] = 'pma__tracking';
$cfg['Servers'][$i]['userconfig'] = 'pma__userconfig';
                              [ Read 61 lines ]
^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify  ^C Cur Pos  ^Y Prev Page  M-\ First Line
^X Exit      ^R Read File  ^\ Replace   ^U Uncut Text ^T To Spell ^_ Go To Line ^V Next Page  M-/ Last Line
```

Ubicamos la línea con respecto al password y le agregamos la clave colocada en la sentencia SQL.

```
/* Authentication type and info */
$cfg['Servers'][$i]['auth_type'] = 'config';
$cfg['Servers'][$i]['user'] = 'root';
$cfg['Servers'][$i]['password'] = '1234';
$cfg['Servers'][$i]['extension'] = 'mysqli';
$cfg['Servers'][$i]['AllowNoPassword'] = true;
$cfg['Lang'] = '';

/* Bind to the localhost ipv4 address and tcp */
$cfg['Servers'][$i]['host'] = '127.0.0.1';
$cfg['Servers'][$i]['connect_type'] = 'tcp';
```

Comprobamos nuevamente. Asegurar tanto para la ip como dominio localhost estén con la clave.




14. Ubicamos el archivo de configuración del plugin mysql.

```
[.]                     [..]                    acl.conf.paho           emqx_auth_http.conf     emqx_auth_jwt.conf
emqx_auth_ldap.conf     emqx_auth_mnesia.conf   emqx_auth_mongo.conf    emqx_auth_mysql.conf    emqx_auth_pgsql.conf
emqx_auth_redis.conf    emqx_bridge_mqtt.conf   emqx_coap.conf          emqx_dashboard.conf     emqx_exhook.conf
emqx_exproto.conf       emqx_lua_hook.conf      emqx_lwm2m.conf         emqx_management.conf    emqx_prometheus.conf
emqx_psk_file.conf      emqx_recon.conf         emqx_retainer.conf      emqx_rule_engine.conf   emqx_sasl.conf
emqx_sn.conf            emqx_stomp.conf         emqx_telemetry.conf     emqx_web_hook.conf
              27 File(s)         68,512 bytes
               2 Dir(s)  29,348,581,376 bytes free

remoto@WIN-HU5I3SRHA93 C:\emqx\etc\plugins>
```

```
 GNU nano 2.5.3                        File: emqx_auth_mysql.conf

##--------------------------------------------------------------------
## MySQL Auth/ACL Plugin
##--------------------------------------------------------------------

## MySQL server address.
##
## Value: Port | IP:Port
##
## Examples: 3306, 127.0.0.1:3306, localhost:3306
auth.mysql.server = 127.0.0.1:3306

## MySQL pool size.
##
## Value: Number
auth.mysql.pool = 8

## MySQL username.
##
## Value: String
#auth.mysql.username =

## MySQL password.
##
## Value: String
#auth.mysql.password =

## MySQL database.
##
## Value: String
auth.mysql.database = mqtt

## MySQL query timeout
##
## Value: Duration
## auth.mysql.query_timeout = 5s

## Variables: %u = username, %c = clientid

## Authentication query.
##
## Note that column names should be 'password' and 'salt' (if used).
## In case column names differ in your DB - please use aliases,
## e.g. "my_column_name as password".
##
## Value: SQL
                                [ Read 131 lines ]
^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify
^X Exit      ^R Read File  ^\ Replace   ^U Uncut Text ^T To Spell
```

Modificamos las líneas de código referente al usuario, base y clave.

```
  GNU nano 2.5.3                        File: emqx_auth_mysql.conf

##--------------------------------------------------------------------
## MySQL Auth/ACL Plugin
##--------------------------------------------------------------------

## MySQL server address.
##
## Value: Port | IP:Port
##
## Examples: 3306, 127.0.0.1:3306, localhost:3306
auth.mysql.server = 127.0.0.1:3306

## MySQL pool size.
##
## Value: Number
auth.mysql.pool = 8

## MySQL username.
##
## Value: String
auth.mysql.username = root

## MySQL password.
##
## Value: String
auth.mysql.password = 1234

## MySQL database.
##
## Value: String
auth.mysql.database = laboratorio

## MySQL query timeout
##
## Value: Duration
## auth.mysql.query_timeout = 5s

## Variables: %u = username, %c = clientid

## Authentication query.
##
## Note that column names should be 'password' and 'salt' (if used).
## In case column names differ in your DB - please use aliases,
## e.g. "my_column_name as password".
##
## Value: SQL
                                                [ Read 131 lines ]
^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify     ^C Cur Po
^X Exit        ^R Read File   ^\ Replace     ^U Uncut Text  ^T To Spell    ^_ Go To
```

15. Ahora procedemos a dirigirnos al dashboard de emqx para iniciar el plugin.




16. Ahora vamos a editar el publicador y suscriptor.

Administrator: c:\windows\system32\cmd.exe - nano publicacion.php

```
 GNU nano 2.5.3                        File: publicacion.php

<!DOCTYPE html>
<head>
<script src="https://unpkg.com/mqtt/dist/mqtt.min.js"></script>
<script>
const clientId = 'mqttjs_' + Math.random().toString(16).substr(2, 8)

const host = 'ws://192.168.0.14:8093/mqtt'
const options = {
  keepalive: 60,
  clientId: clientId,
  username: 'cliente1',
  password: '1234',
  protocolId: 'MQTT',
  protocolVersion: 4,
  clean: true,
  reconnectPeriod: 1000,
  connectTimeout: 30 * 1000,
  will: {
    topic: 'Mensajeros',
    payload: 'Cerrada.!',
    qos: 0,
    retain: false
  },
}

console.log('mqtt client Listo')
const client = mqtt.connect(host, options)
client.on('error', (err) => {
  console.log('Error de conexion: ', err)
  client.end()
})
client.on('reconnect', () => {
  console.log('Reconectando...')
})

//Publicando
client.publish('testtopic', 'MENSAJE!!!...!', { qos: 0, retain: false })

</script>
</head>
<body></body>
</html>



                [ Read 43 lines (Converted from DOS format) ]
^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify     ^C Cur Pos     ^Y Pr
^X Exit        ^R Read File   ^\ Replace     ^U Uncut Text  ^T To Spell    ^_ Go To Line  ^V Ne
```

Administrator: c:\windows\system32\cmd.exe - nano suscripcion.php

```
 GNU nano 2.5.3                          File: suscripcion.php

ï»¿<!DOCTYPE html>
<head>
<script src="https://unpkg.com/mqtt/dist/mqtt.min.js"></script>
<script>
const clientId = 'mqttjs_' + Math.random().toString(16).substr(2, 8)
const host = 'ws://192.168.0.14:8093/mqtt'
const options = {
  keepalive: 60,
  clientId: clientId,
  username: 'cliente1',
  password: '1234',
  protocolId: 'MQTT',
  protocolVersion: 4,
  clean: true,
  reconnectPeriod: 1000,
  connectTimeout: 30 * 1000,
  will: {
    topic: 'Mensajeros',
    payload: 'Cerrada.!',
    qos: 0,
    retain: false
  },
}
console.log('mqtt client Listo')
const client = mqtt.connect(host, options)

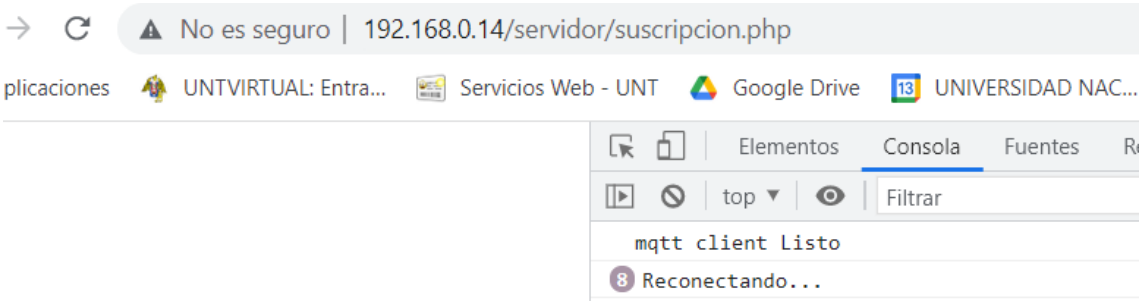
client.on('error', (err) => {
  console.log('Error de conexion: ', err)
  client.end()
})

client.on('reconnect', () => {
  console.log('Reconectando...')
})

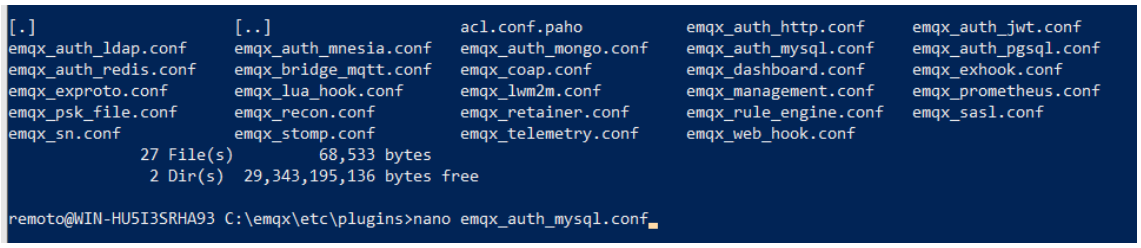

client.on('connect', () => {
  console.log('Cliente Listo:' + clientId)
  // Subscribe
  client.subscribe('testtopic', { qos: 0 })
})

// Recibiendo datos
client.on('message', (topic, message, packet) => {
  console.log('RecepciÃ³n de Mensaje: ' + message.toString() + '\ndel Topico:
                                [ Read 52 lines (Converted from DOS form
^G Get Help    ^O Write Out   ^W Where Is    ^K Cut Text    ^J Justify     ^C
^X Exit        ^R Read File   ^\ Replace     ^U Uncut Text  ^T To Spell    ^_
```

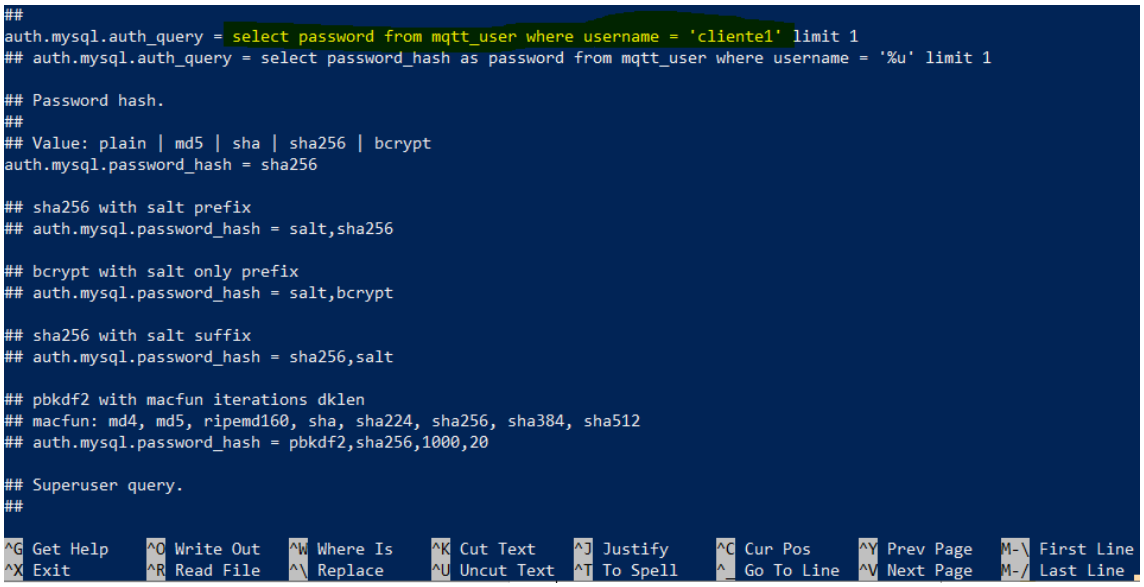
Luego ejecutamos y tenemos problemas de conexión. Por tanto debemos acomodar nuestros datos en la configuración del plugin.



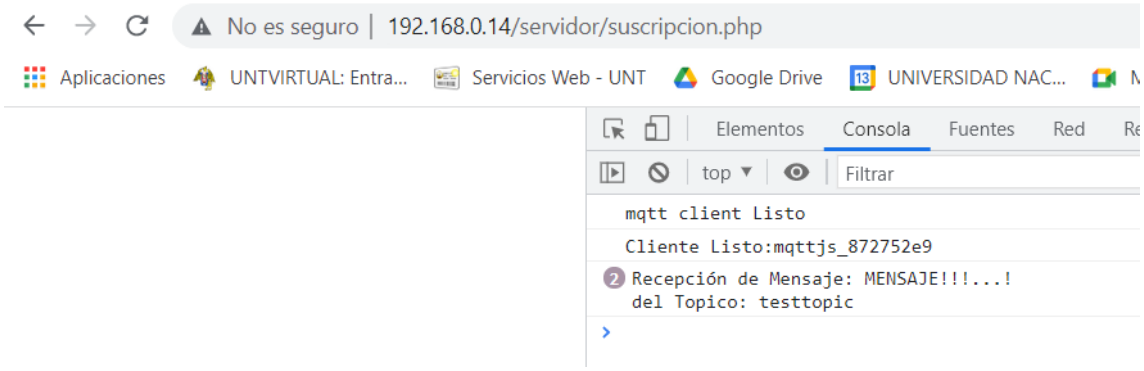

17. Ubicamos el archivo de configuración.

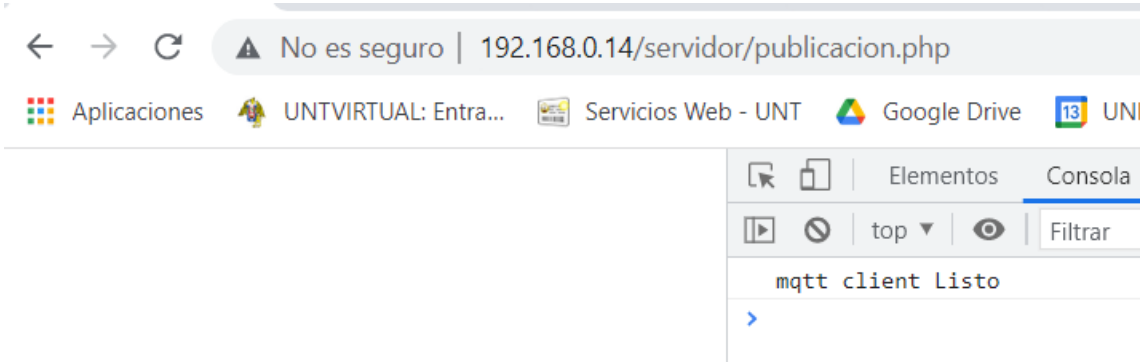



Editamos el SQL para agregar el cliente de la tabla.




18. Comprobamos nuevamente la ejecución de la publicación y suscripción. Todo estará conforme.




No es seguro | 192.168.0.14/servidor/publicacion.php
Aplicaciones
UNTVIRTUAL: Entra...
Servicios Web - UNT
Google Drive
Elementos
Consola
top
Filtrar
mqtt client Listo